

LIKHITH KUMAR MASURA

Forward-Deployed AI Engineer · GenAI · LLM Systems · Backend
Hyderabad, India | +91 85010 69530 | likhithmasura@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

SUMMARY

Forward-deployed AI engineer who ships production LLM systems for enterprise clients — multi-tenant conversational commerce, market-intelligence pipelines, and multi-channel content generation — owning delivery from prompt design through integration with client systems to live rollout. Works across the stack: agentic workflows and RAG on FastAPI backends, provider-agnostic model routing with per-tenant cost attribution, retrieval grounding and guardrails to keep model output truthful, and deployment on AWS.

EDUCATION

Vellore Institute of Technology | M.Tech, Computer Science (ML & AI) 2024 – 2026 · CGPA 9.55 / 10
JNTU Hyderabad | B.Tech, Computer Science (AI & ML) 2020 – 2024 · CGPA 7.99 / 10

EXPERIENCE

ApexNeural | Forward-Deployed AI Engineer Hyderabad, India · Aug 2025 – Present

- Deliver production AI systems for enterprise clients across retail, conversational commerce and market research, owning the path from requirement discovery and prompt design through integration with client systems to live rollout.
- Work test-first on the Python agent services — pytest and pytest-asyncio suites behind a 60% coverage gate enforced in CI, with a failing test reproducing every bug before the fix lands as a permanent regression guard.
- Built LLM cost and reliability observability with LiteLLM and Langfuse — per-call, per-tenant and per-model spend attribution, conversation-level latency and resolution rollups, and structured failure traces queryable by cause.
- Designed a provider-agnostic LLM layer that swaps Anthropic Claude and OpenRouter models by environment variable, with circuit breakers that degrade to rule-based NLU rather than hang on a provider failure.

PROJECTS

Multi-Tenant Commerce Platform — omnichannel AI support & conversational commerce FastAPI · Claude · LiteLLM · Postgres RLS · BullMQ · AWS

- Built agent capabilities for a platform serving four retail brands from a single deployment across WhatsApp, Telegram, Instagram and web — product discovery, stock-checked ordering, payment links and order tracking, with automatic escalation to human agents.
- Kept model output truthful by grounding every product claim against the live catalogue behind a pre-send response validator, and added two-stage retrieval — deterministic SKU/barcode lookup with vector-similarity fallback — so most queries never reach the expensive path — on a curated 58-prompt fixture of common queries, 48 resolved with no model call at all.
- Delivered multilingual handling across English, Hindi, Hinglish, Telugu and Tenglish with language mirroring, plus vision-LLM image search with OCR so customers can send a photo instead of a description.
- Contributed to tenant isolation via PostgreSQL row-level security with transaction-scoped context, and to async processing across 9 BullMQ queues with idempotent, backoff-retried handlers.

DBaaS — pain-point mining to market-gap engine FastAPI · Claude 3.5 Sonnet · OpenRouter · async Python

- Built an AI backend that mines unstructured text and Reddit discussions for customer pain points, then ranks market gaps and expands business ideas against established strategic frameworks.
- Exposed it as modular REST services with agent-based routing and Pydantic validation enforced at the ingest boundary, where a malformed post re-raises rather than reaching the LLM stages.

Content Factory — GenOps for multi-channel campaigns FastAPI · RAG · n8n · OAuth 2.0

- Contributed to an end-to-end client build that turns a single prompt into channel-ready images, captions and a posting schedule, taken from product brief through QA to release.
- Wired brand-voice RAG for on-tone generation and a human-in-the-loop approval loop with OAuth 2.0 publishing to the client channels.

CORE TECHNOLOGIES

Languages	Python, TypeScript, JavaScript, SQL
AI / ML	LLM application development, RAG, agentic workflows, prompt & context engineering, multi-intent classification, retrieval grounding & guardrails, embeddings & vector search, Vision LLMs & OCR, LLM evaluation
LLM Tooling	Anthropic Claude, OpenAI, OpenRouter, LiteLLM, Langfuse, EasyOCR / Tesseract
Backend	FastAPI, Pydantic v2, NestJS, REST APIs, async workers, webhook ingestion, JWT / OAuth2, RBAC, SQLAlchemy
Data & Storage	PostgreSQL (row-level security), Redis, BullMQ, vector embeddings
Cloud & DevOps	AWS (ECS Fargate, RDS, ElastiCache, S3, CloudFront), Docker, Terraform, CI/CD (Jenkins · GitHub Actions), Git
Testing	pytest, Jest, Playwright, k6, test-driven development

CERTIFICATIONS

Problem Solving (Intermediate) — HackerRank · Python Essentials — Cisco · TCS iON Career Edge, Young Professional — TCS